PRODUCT BROCHURE

# PRODUCT ROADMAP

## Technology Brochure

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TOBE. ENERGY

PRODUCT GUIDE

# ELECTROLYZER TECHNOLOGY SOLUTIONS

## SUPERIOR EFFICIENCY & COST SAVINGS

Experience unparalleled 94.7% efficiency that drives down energy consumption and operational costs, allowing you to achieve green hydrogen production at an exceptionally competitive price point.

## IMMEDIATE STARTUP & SEAMLESS SCALABILITY

Begin producing hydrogen within seconds and easily scale from pilot projects to large-scale installations. Reduce downtime, adapt to fluctuating demand, and integrate effortlessly with renewable energy sources.

# Empower Your Energy Future

**Tobe Energy's groundbreaking electrolyzers transform how green hydrogen is produced. By achieving an industry-leading 94.7% wall-to-plug efficiency, our technology dramatically lowers energy costs and scales effortlessly—helping you turn clean hydrogen into a practical, reliable, and profitable energy solution.**

## Key Differentiators

**Immediate Startup:**
No lengthy warm-up times. Begin producing hydrogen in seconds—ideal for pairing with intermittent renewable sources, enabling dynamic load balancing and rapid response to market demands.

**Exceptional Efficiency, Lower Costs:**
Operating at near-ambient temperatures and free of exotic materials, our systems minimize both CAPEX and OPEX. The result? More affordable green hydrogen and an accelerated return on investment.

**Sustainable and Scalable:**
By eliminating waste heat and reliance on costly membranes, Tobe Energy's design ensures long-lasting performance. Our modular approach makes it easy to scale from pilot installations to full industrial setups without compromising quality or efficiency.

**Membrane-Free for Maximum Reliability:**
With no fragile membranes or caustic electrolytes, maintenance is simpler, safer, and more predictable. Our systems use standard stainless steel components, driving down costs and complexity.

## How We Compare

**Tobe Energy:** ~35.2 kWh/kg at 94.7% efficiency
**Typical PEM Electrolyzer (e.g., Nel Hydrogen):** ~53.2 kWh/kg

**The Bottom Line:**
**By choosing Tobe Energy's electrolyzer over a typical PEM system, a plant producing 2,500 kg of hydrogen per day could save approximately $4,500 per day in electricity costs (based on $0.10 per kWh).**

## Core Product Line

Find the perfect fit for your application—from small-scale demonstrations to large-scale hydrogen hubs. All Tobe Energy models maintain exceptional efficiency and immediate responsiveness.

| Model | Daily $H_2$ Output (kg/day) | Energy Use (kWh/kg)* | Daily Energy (kWh/day) | Approx. Electrical Rating (kW)** |
|---|---|---|---|---|
| **Tobe-25** | 25 | ~35.2 | ~880 | ~37 |
| **Tobe-250** | 250 | ~35.2 | ~8,800 | ~367 |
| **Tobe-2500** | 2,500 | ~35.2 | ~88,000 | ~3,667 (3.7 MW) |

*Calculated from a 94.7% efficiency using the lower heating value (LHV) of hydrogen (~33.3 kWh/kg).
**Assuming continuous operation.

## Tobe Energy Electrolyzer – Specifications (2,500 kg/day model)

| Category | Tobe Energy Specification |
|---|---|
| **Net Production Rate** | **~1,160 Nm³/h (at 0°C, 1 bar)**<br>**2,500 kg/24 h** |
| **Turndown Range** | **10 to 100%** |
| **Power at Stack at 100% Capacity*** | **~3.16 kWh/Nm³ – ~35.2 kWh/kg** |
| **Purity** | **99+%, saturated gas with liquid water** |
| **$O_2$ Content in $H_2$** | **< 1 ppm v** |
| **$N_2$ Content in $H_2$** | **< 2 ppm v** |
| **Operating Pressure** | **Up to 350 psig (can operate lower)** |
| **Dimensions (W x D x H)** | **~10 m x 2.5 m x 2.5 m (modular design)** |
| **Ambient Temperature Range** | **-30 to 50°C** |
| **Electrolyte** | **Membrane-free, caustic-free** |

*Based on 94.7% efficiency using the lower heating value of hydrogen (~33.3 kWh/kg).

## Membrane-Free Technology

Conventional electrolyzers rely on fragile and costly membranes that drive up costs and limit system life. Tobe Energy eliminates this bottleneck with a fully membrane-free design:

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|---|---|
| **No Caustic Electrolytes** | Eliminates hazards, reduces complexity, and simplifies maintenance. |
| **Longer Lifespans** | Removing the primary failure point (membranes) significantly extends operational life and lowers replacement costs. |
| **Cost-Effective Materials** | Standard 304 stainless steel construction replaces exotic, expensive materials for universal availability and lower CAPEX. |

The result is a solution that's easier to deploy, maintain, and scale, opening the door to hydrogen production in emerging markets and remote regions.

## Versatile Applications

Tobe Energy electrolyzers empower a wide range of sectors, driving innovation and efficiency wherever hydrogen can play a role:

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|---|---|
| **Agriculture & Chemicals** | Ammonia Production, Fertilizer Production, Chemical Feedstock, Desulfurization, Hydrogenation, Methanol & Biomethanol Production |
| **Metallurgy & Materials** | Annealing, Blast Furnace Injection, Direct Reduced Iron, Powder Metallurgy, Protective Atmosphere, Sintering, Catalytic Atmosphere, Float/Flat Glass Production |
| **Energy & Fuel Sectors** | Autoconsumption, Biofuel Production, Biogas/Biomethane Production, Carbon Capture, Energy Storage, Waste-to-Synthetic Fuels |
| **Industrial & Transportation** | Cylinder/Tube Truck Filling, Fuel for Fuel Cells, Fuel for Turbines, Fuel for Hydrogen Internal Combustion Engines (HICE), Extreme Ultraviolet Lithography, Combustion, Grid Stabilization, Space Heating |
| **Advanced Processes** | Direct Gas Injection, Powder Processing |

With Tobe Energy, hydrogen production becomes more accessible, economical, and easily integrated into existing and emerging markets.

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# Redefining Cost and Efficiency

## Overcoming Traditional Shortcomings

Legacy systems struggle with waste heat, reliance on exotic materials, and high-pressure, high-temperature conditions. Tobe Energy's approach addresses each of these drawbacks:

| | |
|---|---|
| **No Waste Heat, No Cooling** | Eliminates the need for complex heat management, reducing both CAPEX and OPEX. |
| **Standard Materials** | Avoiding rare metals and membranes drastically cuts component costs. |
| **Immediate Startup** | Seamless hydrogen production on-demand, with no hazardous chemicals, enhances safety and operational flexibility. |

These improvements translate directly into lower costs, easier maintenance, and scalable solutions that drive broad adoption of green hydrogen.

## Why It Matters

**Tobe Energy is not just refining existing technology; we are reimagining hydrogen production from the ground up. Our system opens new possibilities for clean, efficient, and economical hydrogen across industries, geographies, and scales.**

## Take the Next Step

Interested in exploring a pilot installation or sizing a system for your specific needs? Our expert team is ready to guide you through integration, financial modeling, and long-term energy strategies.

## Contact Us

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**Join us in shaping a cleaner, more sustainable future—together, we can revolutionize the global energy landscape with accessible, high-performance green hydrogen.**